25. The equation of directrix of the parabola $y^2 = -x$ is

(a) $4x + 1 = 0$ (b) $4x - 1 = 0$
(c) $x - 4 = 0$ (d) $x + 4 = 0$

Miscellaneous Problems

1. A point P moves so that P and the points (2, 2) and (1, 5) are always collinear. Find the locus of P.
2. As the number of units produced increases from 500 to 1000 and the total cost of production increases from ₹ 6000 to ₹9000. Find the relationship between the cost (y) and the number of units produced (x) if the relationship is linear.
3. Prove that the lines $4x + 3y = 10$, $3x - 4y = -5$ and $5x + y = 7$ are concurrent.
4. Find the value of p for which the straight lines $8px + (2 - 3p)y + 1 = 0$ and $px + 8y - 7 = 0$ are perpendicular to each other.
5. If the slope of one of the straight lines $ax^2 + 2hxy + by^2 = 0$ is thrice that of the other, then show that $3h^2 = 4ab$.
6. Find the values of a and b if the equation $(a-1)x^2 + by^2 + (b-8)xy + 4x + 4y - 1 = 0$ represents a circle.
7. Find whether the points $(-1, -2)$, $(1, 0)$ and $(-3, -4)$ lie above, below or on the line $3x + 2y + 7 = 0$.
8. If $(4, 1)$ is one extremity of a diameter of the circle $x^2 + y^2 - 2x + 6y - 15 = 0$, find the other extremity.
9. Find the equation of the parabola which is symmetrical about x-axis and passing through (–2, –3).
10. Find the axis, vertex, focus, equation of directrix and the length of latus rectum of the parabola $(y-2)^2 = 4(x-1)$

Summary

- Angle between the two intersecting lines $y = m_1x + c_1$ and $y = m_2x + c_2$ is $\theta = \tan^{-1}\left[\left|\frac{m_1 - m_2}{1 + m_1 m_2}\right|\right]$.
- Condition for the three straight lines $a_1x + b_1y + c_1 = 0$, $a_2x + b_2y + c_2 = 0$, and $a_3x + b_3y + c_3 = 0$ to be concurrent is $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0$.
- The condition for a general second degree equation $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ to represent a pair of straight lines is $abc + 2fgh - af^2 - bg^2 - ch^2 = 0$.
- Pair of straight lines passing through the origin is $ax^2 + 2hxy + by^2 = 0$.
- Let $ax^2 + 2hxy + by^2 = 0$ be the straight lines passing through the origin then the product of the slopes is $\frac{a}{b}$ and sum of the slopes is $\frac{-2h}{b}$.
- If θ is the angle between the pair of straight lines. $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$, then $\theta = \tan^{-1}\left[\left|\frac{\pm 2\sqrt{h^2 - ab}}{a + b}\right|\right]$.

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